

Michael Zandonella

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EDUCATION

Oregon State University

Bachelor of Science in Computer Science | 3.96 GPA

Corvallis, OR

Expected June 2026

EXPERIENCE

Computer Science Peer Tutor

October 2025 – Present

College of Engineering, Oregon State University

Corvallis, OR

- Solved complex debugging, data structures, and algorithm problems under time constraints by rapidly adapting to unfamiliar codebases, identifying edge cases, and guiding software implementations
- Improved student understanding in Python, React, SQL, HTML, CSS, JavaScript, networking, security, and algorithmic problem solving by delivering targeted 1-on-1 technical instruction to 5-15 students per term

Web Assistant

June 2024 - July 2025

OSU Beaver Store

Corvallis, OR

- Reduced back stock item retrieval time by 55% by building an internal Python automation tool that scraped product images and catalog data to generate printable inventory labels
- Maintained data quality and improved catalog consistency across thousands of e-commerce listings and 500+ product assets by auditing catalog data and implementing data-driven content improvements

PROJECTS

Rotations.lol | TypeScript, React, Node.js, Supabase, PostgreSQL, AWS SES, Docker

- Enabled 500+ users to monitor high-value discounts across 16,000+ in-game items with 99.9% platform uptime, helping users save an average of 46% off retail pricing by building and operating a full-stack sale tracking platform with live data ingestion and automated notifications
- Maintained continuously updated pricing and catalog data across ~80 new weekly sales by engineering a distributed ingestion pipeline that processes and stores live sale data in PostgreSQL
- Delivered reliable automated sale notifications across 6,100+ user-wishlisted items by developing a scheduled event-processing pipeline that matches live pricing updates, suppresses duplicate alerts, and automates email delivery through AWS SES and SNS

RFID-Based Asset Tracking System (Senior Capstone) | Node.js, React, Socket.IO, Express, SQLite, Docker

- Established real-time asset visibility and role-based asset tracking workflows for 400+ students and staff by leading integration across frontend, backend, and RFID scanner systems, architecting Socket.IO event contracts, and collaborating with stakeholders to refine operational requirements
- Achieved sub-second synchronization between RFID scanner hardware and a multi-user frontend by architecting an event-driven asset tracking system with resilient device recovery and a custom scanner SDK
- Accelerated development and stakeholder testing for a 5-person engineering team by building a mock scanner API and control interface that simulated RFID hardware interactions during 6 months of hardware unavailability

YouTube Transcript Search | Python, Elasticsearch, React, Flask

- Implemented timestamp-level search across arbitrary YouTube playlists with indexing throughput benchmarked at 4,400+ transcript segments per second by designing a full-stack retrieval system using Elasticsearch
- Improved transcript search precision and recall to a 0.955 NDCG@10 score across 15 evaluated search queries by implementing BM25-based ranking, exact phrase matching, fuzzy search, and multi-field query boosting
- Reduced long-form video search and navigation time from minutes to seconds across 250+ video playlists by building a scalable React and Flask transcript search platform with direct timestamped playback links

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS

Frontend & Backend: React, Node.js, Express, Flask, REST APIs, WebSockets

Databases & Search: PostgreSQL, MongoDB, Elasticsearch, Supabase

Infrastructure & Tools: Docker, CI/CD, AWS, Nginx, Linux, Git